

Functional Equations

Note $\forall$ means "for all", $\mathbb{R}$ is the Real Numbers, $\mathbb{R}^+$ is the set of positive Reals.

Question difficulty is indicated by 🐣 (lower difficulty), 🐣🐣 (greater difficulty), and 🐣🐣🐣 (greatest difficulty). A topic label is also given on the question.

Rating: 🐣

Problem 1

Solve $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x+y)^2 = f(x)^2 + f(y)^2$, $\forall x, y \in \mathbb{R}$

Problem 2

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x^2 + y) = f(x^{27} + 2y) + f(x^4)$.

Problem 3

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f\left(\frac{x+f(x)}{2} + y\right) = f(x) + y$.

Problem 4

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $x^2 f(x) + f(1-x) = 2x - x^4$.

Problem 5

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(x) + y^2 + f(xy) = f(x+y)^2 - f(x)f(y)$$

for all real numbers x, y .

Rating: 🐣🐣

Problem 6

(USAMO 2002) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x^2 - y^2) = xf(x) - yf(y)$.

Problem 7

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all positive integers x, y we have

$$2yf(f(x^2) + x) = f(x+1)f(2xy).$$

Problem 8

Let $\mathbb{R}^+$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x^2 + xf(y)) = f(f(x))(x + y)$$

for all positive real numbers x and y .

Problem 9

(SMMC 2024 B2): Find all continuous functions $f : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ such that

$$f(x) = (x + 1)f(x^2),$$

for all $x \in \mathbb{R} \setminus \{-1, 1\}$.

Problem 10

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(y)) + f(xy) = f(x + 1)f(y + 1) - 1.$$

Rating: 🐣🐣🐣

Problem 11

Let $\mathbb{R}^+$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)(f(f(x^2)) + f(f(y^2)))$$

for all positive real numbers x and y .

Problem 12

(Putnam 2010 B5) Determine whether there exists a strictly increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the differential equation

$$f'(x) = f(f(x)) \quad \text{for all } x \in \mathbb{R}.$$